

La'Tahvia (Tahvia) Williams

Data Analyst Intern · Python · SQL · Statistical Modeling

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SUMMARY

Data Science and Public Policy student at the University of Chicago with production experience in Python, SQL, and applied statistics. Led a 3-person team modeling ~300,000 federal sentencing records and built a pipeline processing \$157.8M across 1.18M pharmaceutical payment transactions. Ship end-to-end, from data ingestion to deployed application.

EDUCATION

University of Chicago

B.A., Data Science and Public Policy (Double Major) GPA: 3.3

Relevant coursework: Statistical Models and Regression, Machine Learning, Data Engineering, Big Data, Data Visualization

Chicago, IL

Expected 2028

TECHNICAL SKILLS

Languages: Python, SQL, R, TypeScript, JavaScript

Analysis & ML: pandas, NumPy, scikit-learn, statsmodels, OLS / logistic / LASSO regression, fixed effects, hypothesis testing

Visualization: Altair, Plotly, matplotlib, seaborn, ggplot2, D3, Chart.js

Data & Systems: PostgreSQL, FastAPI, Chroma (RAG), REST APIs, React, Git, GitHub Actions, Railway, Vercel

EXPERIENCE

Rights Within Reach

Founder & Developer

Chicago, IL

Jun. 2026 – Present

- Built and deployed rightswithinreach.org, a multilingual legal-rights retrieval app indexing 130+ documents across 5 languages for users; FastAPI/Chroma RAG backend, React/TypeScript frontend, shipped on Railway and Vercel.
- Designed the retrieval architecture with metadata pre-filtering, hybrid BM25 search, and cross-encoder reranking, improving retrieval accuracy 89% over a dense-only baseline.

TectoniQ

Quantitative Research Fellow

Remote

Jun. 2026 – Aug. 2026

- Benchmarked a quantum fidelity kernel against tuned classical baselines, running a model ladder from logistic regression to gradient boosting under equal-tuning protocols.
- Overtuned three prior over-optimistic findings by rebuilding the evaluation with bootstrap confidence intervals and DeLong testing, establishing no significant edge over gradient boosting.
- Delivered a 15-page research paper and reproducibility runbook; verified results across the Quantinuum toolchain (Guppy, pytket, Nexus) and modeled hardware cost to scale the kernel matrix.

Spry Digital

Technology Intern

St. Louis, MO

Jun. 2025 – Aug. 2025

- Audited 10+ client websites for UX and accessibility, delivering recommendations that fed into 5+ digital transformation projects.
- Built REST API integrations and custom GPT workflows automating internal reporting and large-scale data migration.

Armanino LLP

Auditing Intern

St. Louis, MO

Jun. 2024

- Audited 10+ Fortune 500 401(k) plans and ran contract competency reviews across 10 programs at 99% compliance.

Reinsurance Group of America

Enterprise Program Management Intern

St. Louis, MO

Jun. 2023 – Aug. 2023

- Rebuilt a training-material management system for a 100+ staff library, improving content accuracy 30%; coordinated 25+ leadership meetings.

PROJECTS

Sentenced Differently — Disparities in Federal Sentencing

Python, statsmodels, pyreadstat, Altair

- Led a 3-person team as Project Manager and Tech Lead, analyzing ~300,000 federal sentencing records (FY2020–FY2024) for residual race and gender disparities after legal controls.
- Built a two-stage model separating the incarceration decision (logistic) from sentence length (log-linear OLS) with district and fiscal-year fixed effects; validated with VIF, Cook's distance, and robust SEs.

Open Payments Analysis — Pharmaceutical Industry Transparency

Python, pandas, Plotly, CMS data

- Analyzed \$157.8M across 1.18M payment transactions to OB/GYN specialists (2017–2020); identified a 43% pandemic-year collapse (\$46.5M → \$26.3M) and a 140% rise in average Reproductive Endocrinology payments.

The Geography of Rights — 50-State Policy Index

Python, statsmodels, Altair

- Scored all 50 states on a composite Human Rights Policy Index across five policy domains using PCA weighting; modeled political and demographic predictors via OLS.

PublicPath — Public-Sector Job & Partner Hub

Python, PostgreSQL, GitHub Actions

- Built Python sync scripts feeding a PostgreSQL backend for a jobs-and-partnerships platform; automated daily ingestion via GitHub Actions.

LEADERSHIP & ACTIVITIES

- Inter-House Council, President** (prev. Vice President): Residential governance for 4,000+ students; run a twice-per-quarter funding cycle.
- Resident Assistant, University of Chicago: Support 100+ residents;** design educational and community programming.
- The Chicago Maroon, Data Reporter:** Financial aid analysis benchmarking UChicago against 9 peer institutions using Common Data Sets.
- Greenwood Project, FinTech Scholar:** A selective program placing students into quantitative and financial services roles.